

Exam Date & Time: 02-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination, Jan -2025

Ethical Hacking and Cyber Security [CSE-4066]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the core objectives of computer security, focusing on confidentiality, integrity, and availability (CIA). With an appropriate example, discuss how do these principles help protect information systems. (5)
A)
B) Identify and demonstrate the command to be used in fping tool to perform the following task
a. To check whether the list of hosts whose ip are stored in a file named selectedhosts.txt are alive (5)
b. ping multiple interfaces on target host
- 2) Examine in detail about cyber attack life cycle. (5)
A)
B) Illustrate in detail about mitigations related to VoIP and IPsec enumerations. (5)
- 3) Demonstrate the common web application vulnerabilities, such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). Explain the impact of these vulnerabilities on application security (5)
A)
B) Discuss the following:
i) ODBC (5)
ii) OLE DB
- 4) Illustrate the various password-cracking techniques, including brute-force attacks, dictionary attacks, and rainbow tables. What are some best practices to strengthen password security? (5)
A)
B) Explain how digital signatures are legally recognized under the Indian IT Act and discuss their role in securing online transactions. (5)
- 5) Examine the common real time cyberattacks on mobile phones, such as spyware, malware, and SMS phishing. Discuss how do these attacks compromise user data? (5)

- A)
- B) Illustrate the importance of the application layer in web applications and explain how vulnerabilities in this layer can lead to security issues. (5)

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